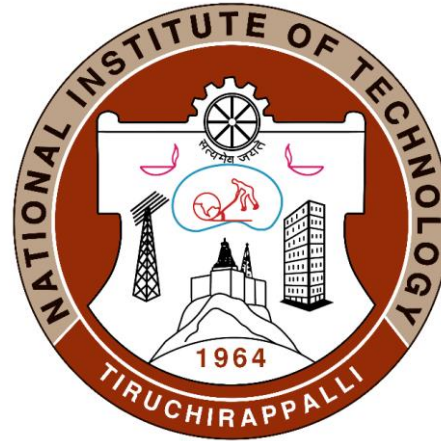


Introduction to Polymers as an Engineering Material



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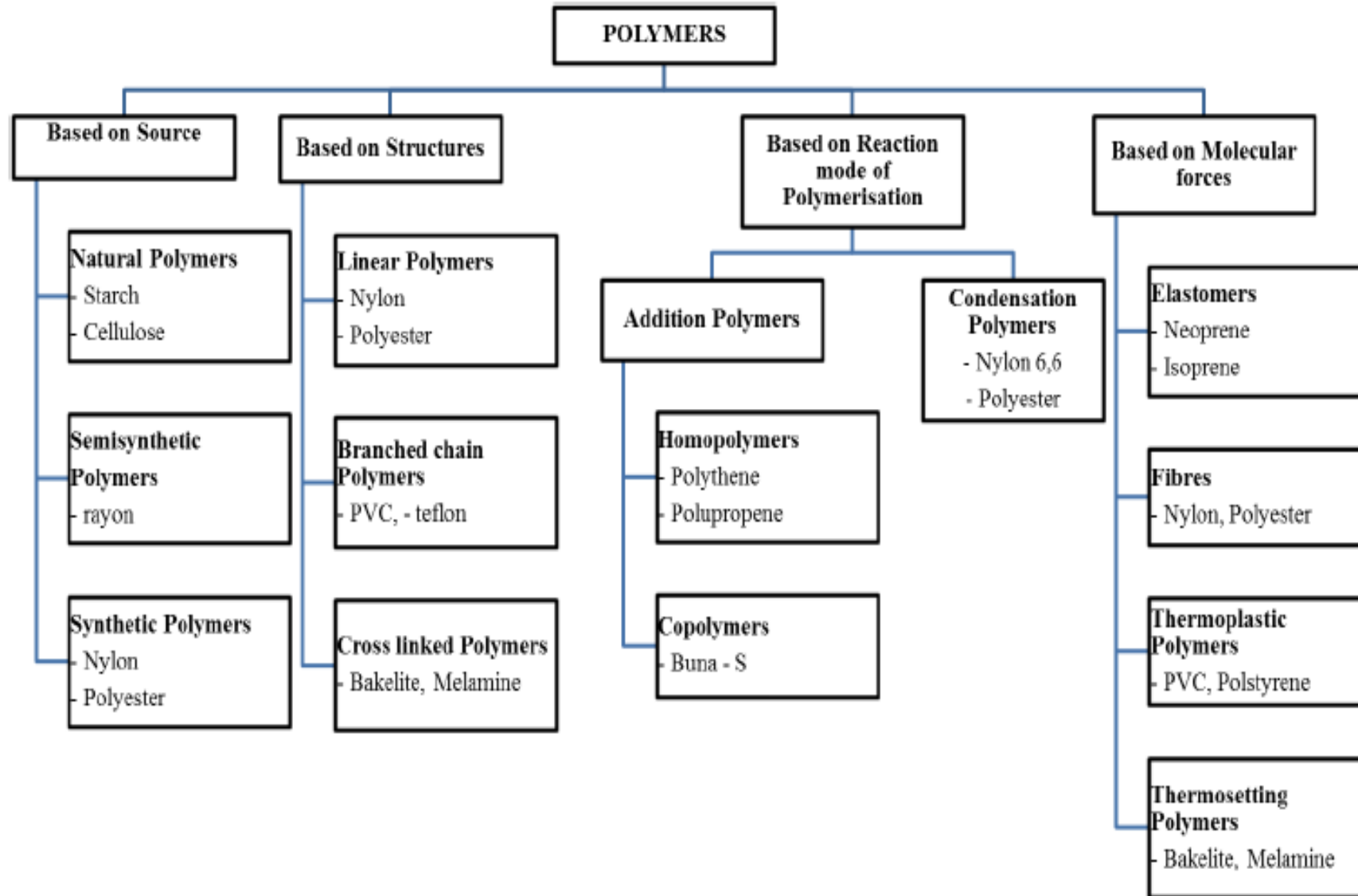
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Lecture 3

Molecular Weight

Broad Classification of Polymers

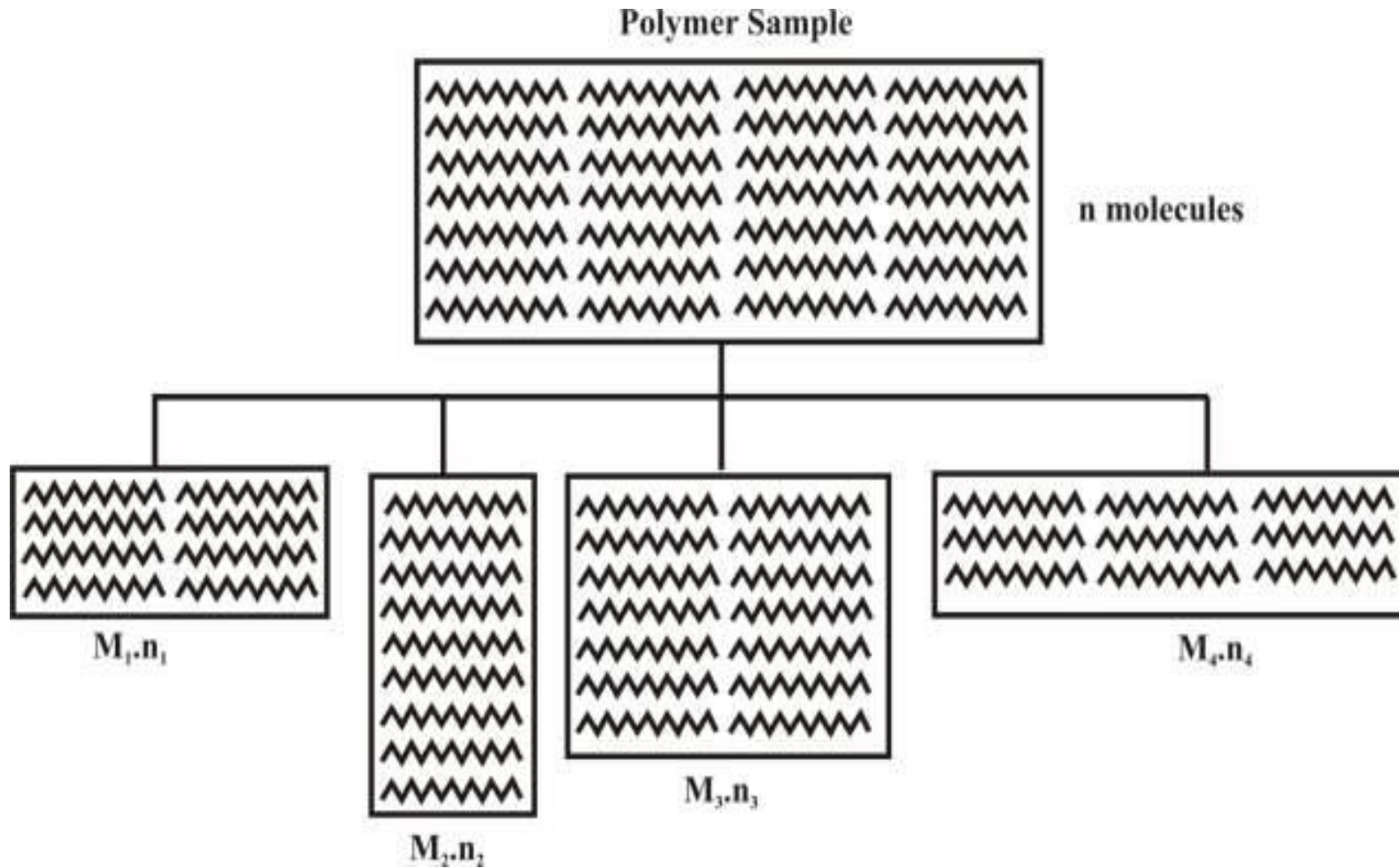


Molecular weight of Polymers

“Molecular weight of a polymer is defined as sum of the atomic weight of each of the atoms in the molecules, which is present in the polymer”.

This distribution of molecular weights is caused by the statistical nature of the polymerization process e.g. methane (CH_4) molecules have the same molecular weight (16), but all polyethylene do not have the same molecular weight because the statistical distribution of molecular weight may be different for the different grade of the polyethylene and the degree of polymerization may also be different.

Generalization concept



Molecular weight

The molecular weight (chain length) is controlled by the synthesis process: Relative rates of initiation, propagation, termination steps of polymerization

Formation of macromolecules during polymerization results in a distribution of chain lengths and molecular weights

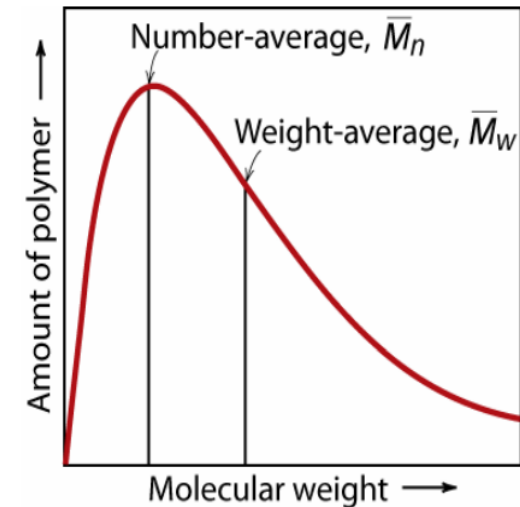
The average molecular weight can be obtained by averaging the masses with the fraction of times they appear (number average molecular weight) or with the mass fraction of the molecules (weight-average molecular weight).

number-average:

$$\bar{M}_n = \sum x_i M_i$$

weight-average:

$$\bar{M}_w = \sum w_i M_i$$



w_i is weight fraction of chains of length i

x_i is number fraction of chains of length i

Molecular weight: Example illustrating the difference between number-average and weight-average

student	weight mass (lb)
1	104
2	116
3	140
4	143
5	180
6	182
7	191
8	220
9	225
10	380

What is the average weight of students in this class

- (a) Based on the number fraction of students in each mass range ?
- (b) Based on the weight fraction of students in each mass range ?

Solution

The first step is to sort the students into weight ranges (let's use 40 lb ranges)

weight range	# of students	mean weight	number fraction	weight fraction
	N_i	M_i	x_i	w_i
81-120	2	110	0.2	0.117
121-160	2	142	0.2	0.150
161-200	3	184	0.3	0.294
201-240	2	223	0.2	0.237
241-280	0	-	0	0.000
281-320	0	-	0	0.000
321-360	0	-	0	0.000
361-400	1	380	0.1	0.202

$$\sum N_i = 10 \quad \sum N_i M_i = 1881 \quad x_i = N_i / \sum N_i \quad w_i = N_i M_i / \sum N_i M_i$$

$$\bar{M}_n = \sum x_i M_i = 0.2 \times 110 + 0.2 \times 142 + 0.3 \times 184 + 0.2 \times 223 + 0.1 \times 380 = 188 \text{ lb}$$

$$\bar{M}_w = \sum w_i M_i = 0.117 \times 110 + 0.150 \times 142 + 0.294 \times 184 + 0.237 \times 223 + 0.202 \times 380 = 218 \text{ lb}$$

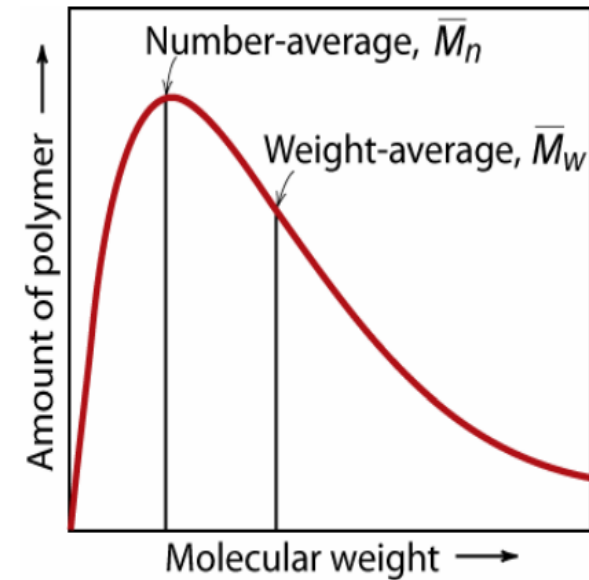
$$\bar{M}_w > \bar{M}_n$$

number-average:

$$\bar{M}_n = \sum x_i M_i$$

weight-average:

$$\bar{M}_w = \sum w_i M_i$$



w_i is weight fraction of chains of length i

x_i is number fraction of chains of length i

Example 2 Consider a polymer, which contains four molecular weight polymers in different numbers and weight and these

Polymer entity	Number of unit in each entity, n	Weight of each grams, M(g)	Total weight of each entity, $W = nM(g)$
Poly-1	2	10	20
Poly-2	4	20	80
Poly-3	6	100	600
Poly-4	3	250	750
Total	15	.	1450

Total number of polymers in containing each entity of poly-1, poly-2, poly-3 and poly-4 is = 15

Number of Poly-1 present in the polymer = 2

Number of fractions of poly-1 = $2/15$

Similarly, Number of fractions of poly-2 = $4/15$, Number of fractions of poly-3 = $6/15$, Number of fractions of poly-4 = $3/15$

Contribution made by poly-1 towards the average weight of polymer = number of fractions of each polymer x weight of each poly entity

Therefore, each poly contribution is

$(2/15) \times 10 = 1.33\text{g}$, $(4/15) \times 20 = 5.33\text{g}$, $(6/15) \times 100 = 40\text{g}$, $(3/15) \times 250 = 50\text{g}$

Summing up the contribution to get Number Average Molecular Weight=

$$1.33 + 5.33 + 40 + 50 = 96.66\text{g}$$

Generalization of Concept

Total number of molecules (n) is given by

$$n = n_1 + n_2 + n_3 + n_4 + \dots = \sum n_i$$

Number fraction of each molecule is = $\frac{n_i}{\sum n_i}$

Number average weight contribution of each entity is = $\frac{n_i M_i}{\sum n_i}$

Number average weight molecular weight is

$$\frac{n_1 M_1}{\sum n_i} + \frac{n_2 M_2}{\sum n_i} + \frac{n_3 M_3}{\sum n_i} + \frac{n_4 M_4}{\sum n_i} + \dots = \frac{\sum n_i M_i}{\sum n_i} = M_n$$

Weight Average Molecular Weight (Mw)

Total weight of each poly present in the polymer =1450g

Weight of poly-1 present in polymer = 20g

Weight fraction of poly-1 = $20/1450$, Weight fraction of poly-2 = $80/1450$, Weight fraction of poly-3 = $600/1450$, Weight fraction of poly-4 = $750/1450$

Contribution made by each poly towards average weight of polymer = weight fraction of poly-1x weight of each unit

For poly-1 $(20/1450) \times 10 = 0.14\text{g}$

For poly-2 $(80/1450) \times 20 = 1.10\text{g}$

For poly-3 $(600/1450) \times 100 = 41.38\text{g}$

For poly-4 $(750/1450) \times 250 = 129.31\text{g}$

Summing up the contribution made by each poly to get weight average molecular weight is

$$0.14 + 1.10 + 41.38 + 129.31 = 171.93\text{g}$$

Generalization of Concept

Total number of molecules (n) is given by

$$n = n_1 + n_2 + n_3 + n_4 + \dots = \sum n_i$$

Total weight of the polymer is = $\sum n_i M_i$ W

Weight fraction of each molecule is = $\frac{n_1 M_1}{W} = \frac{n_1 M_1}{\sum n_i M_i}$

Weight average weight contribution of each entity is = $\frac{n_1 M_1 M_1}{\sum n_i M_i} = \frac{n_1 M_i^2}{\sum n_i M_i}$

Number average weight molecular weight is

$$\frac{n_1 M_1^2}{\sum n_i M_i} + \frac{n_2 M_2^2}{\sum n_i M_i} + \frac{n_3 M_3^2}{\sum n_i M_i} + \frac{n_4 M_4^2}{\sum n_i M_i} + \dots = \frac{\sum n_i M_i^2}{\sum n_i M_i} = M_w$$

For synthetic polymers M_w is greater than the M_n . If they are equal then they will be considered as perfectly homogeneous. (Each molecule has same molecular weight).

Molecular weight and degree of polymerisation

Number of repeating unit in a polymer called as *degree of polymerisation* (DP). DP provides the indirect method of expressing the molecular weight and the relation is as follows;

$$M = DP \times m$$

Where, M is the molecular weight of polymer, DP is the degree of polymerisation and m is the molecular weight of the monomer

$$(DP)_n = \frac{\sum n_i (DP)_i}{\sum n_i} \text{ and } (DP)_w = \frac{\sum n_i (DP)_i^2}{\sum n_i (DP)_i}$$

Each of these averages can be related to the corresponding molecular weight average by the following two equations;

$$M_n = (DP)_n \cdot m$$

$$M_w = (DP)_w \cdot m$$

Degree of polymerization

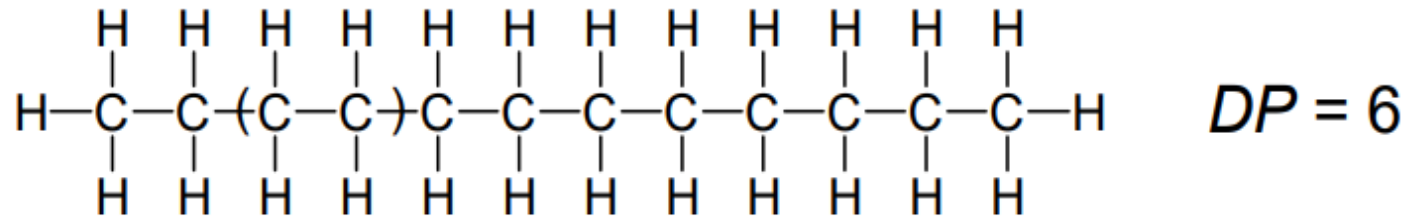
Alternative way to express average polymer chain size is degree of polymerization
- the average number of mer units in a chain

or

Number of repeating unit in a polymer called as *degree of polymerisation* (DP).

$$DP = \frac{\overline{M}_n}{\overline{m}}$$

$\overline{m}$ is the average molecular weight of repeat unit
for copolymers it is calculated as $\overline{m} = \sum f_i m_i$
(f_i is fraction of mer i of molecular weight m_i)



Properties of polymers depend on molecular weight

The increase in the molecular weight leads to increase in

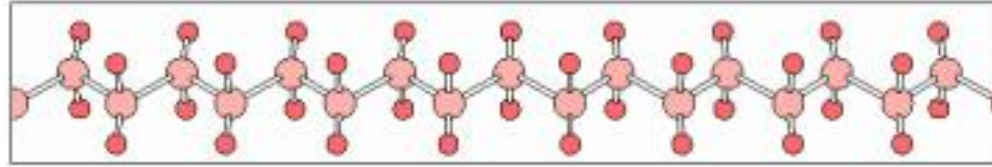
- (i) Melt Viscosity
- (ii) Impact strength

Lowers in

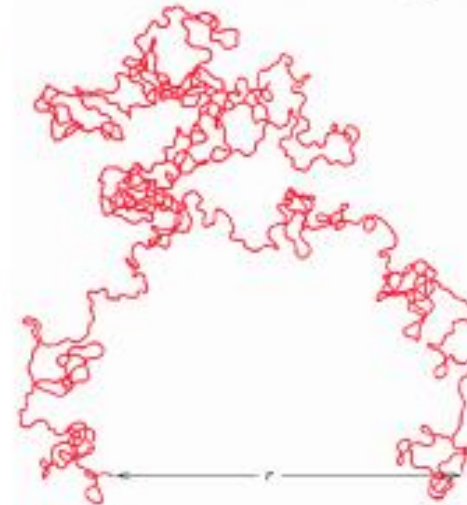
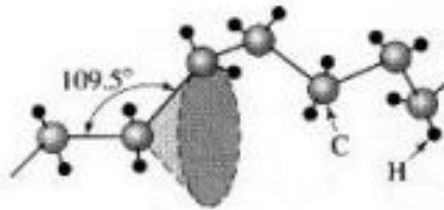
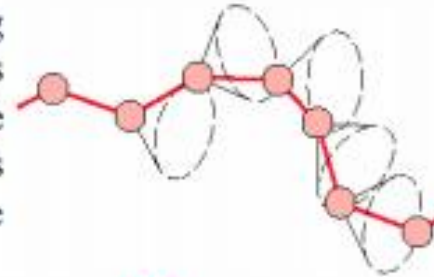
- (i) Hardness
- (ii) Stiffness
- (iii) Softening point
- (iv) Brittle point

High molecular weight polymer does not crystallize so easily as lower molecular weight material crystallizes due to chain entanglement and that reflect in bulk properties of the high molecular weight polymer.

Molecular Shape

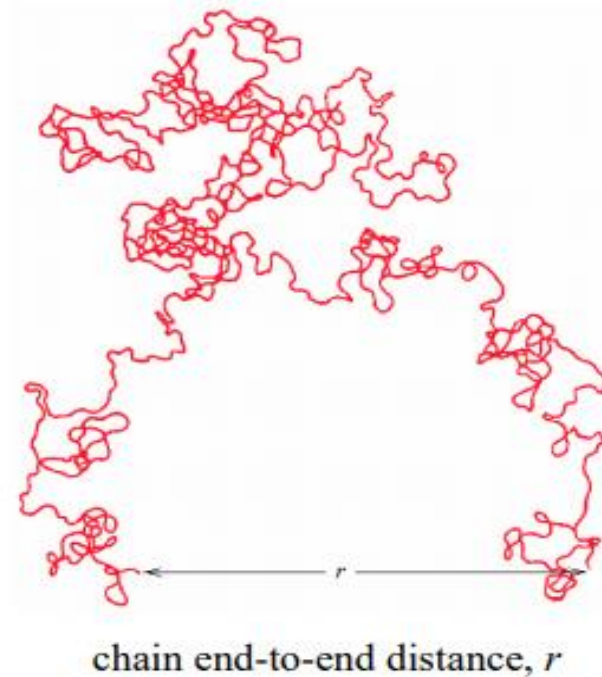


- The angle between the singly bonded carbon atoms is $\sim 109^\circ$ – carbon atoms form a zigzag pattern in a polymer molecule.
- Moreover, while maintaining the 109° angle between bonds polymer chains can rotate around single C-C bonds (double and triple bonds are very rigid).
- Random kinks and coils lead to entanglement, like in the spaghetti structure:



Molecular Shape II

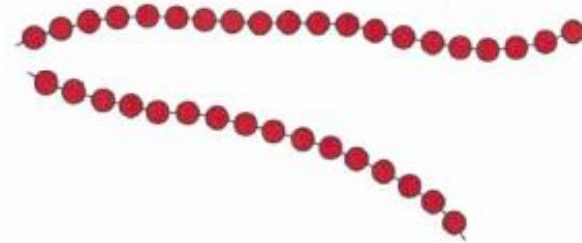
- Molecular chains may thus bend, coil and kink
- Neighboring chains may intertwine and entangle
- Large elastic extensions of rubbers correspond to unraveling of these coiled chains
- Mechanical/thermal characteristics depend on the ability of chain segments to rotate



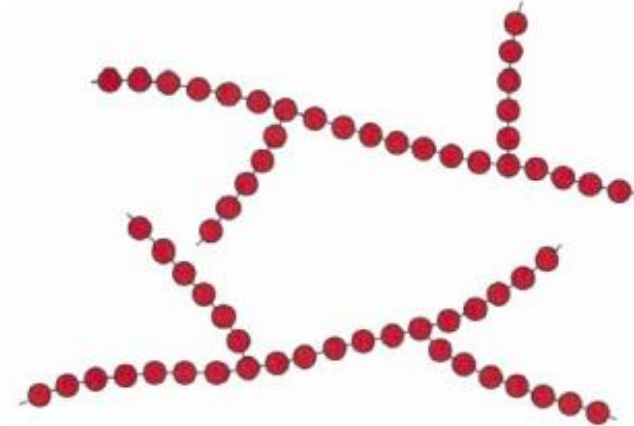
Molecular Structure

The physical characteristics of polymer material depend not only on molecular weight and shape, but also, on molecular structure

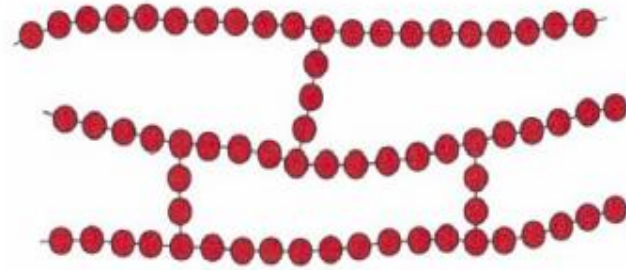
- 1 **Linear polymers:** Van der Waals bonding between chains. Examples: polyethylene, nylon.



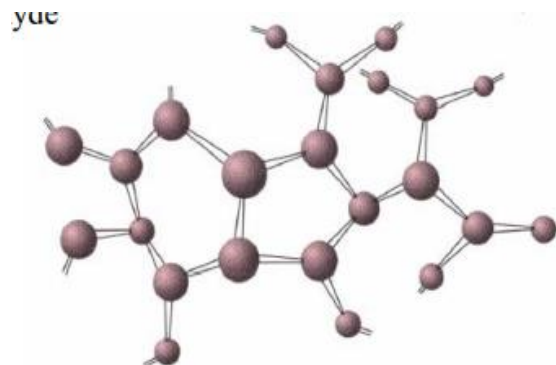
- 2 **Branched polymers:** Chain packing efficiency is reduced compared to linear polymers - lower density



3 Cross-linked polymers: Chains are connected by covalent bonds, often achieved by adding atoms or molecules that form covalent links between chains. Many rubbers have this structure



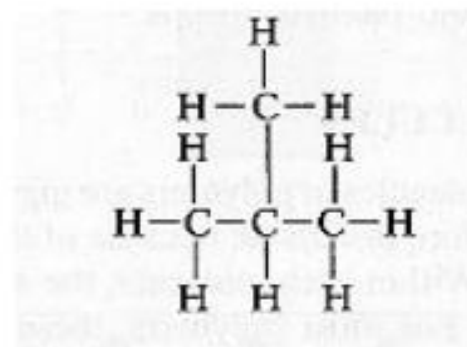
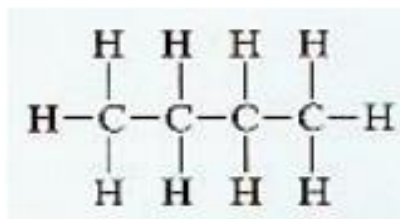
4 Network polymers: 3 D networks made from trifunctional mers. Examples: epoxies, phenol formaldehyde



Isomerism

Isomerism: Hydrocarbon compounds with same composition may have different atomic arrangements.

Physical properties may depend on **isomeric state** (e.g. boiling temperature of normal butane is $-0.5\text{ }^{\circ}\text{C}$, of isobutane $-12.3\text{ }^{\circ}\text{C}$)



Butane $\rightarrow$ C₄H₁₀ $\leftarrow$ Isobutane

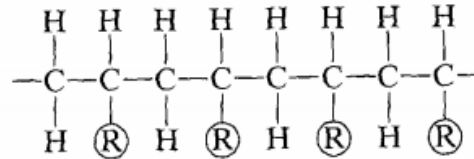
Two types of isomerism in polymers are possible:

- (I) stereoisomerism
- (II) geometrical isomerism

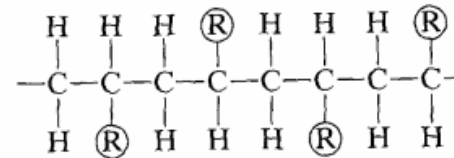
Stereoisomerism

atoms are linked together in the same order, but can have different spatial arrangement

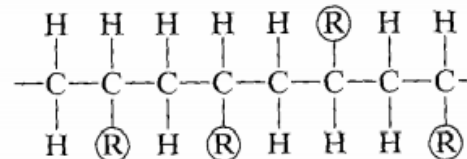
- 1 **Isotactic configuration:** all side groups R are on the same side of the chain.



- 2 **Syndiotactic configuration:** side groups R alternate sides of the chain.

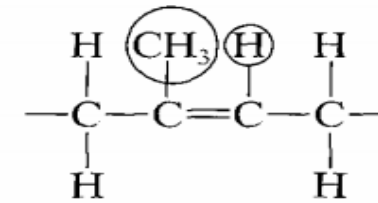


- 3 **Atactic configuration:** random orientations of groups R along the chain.

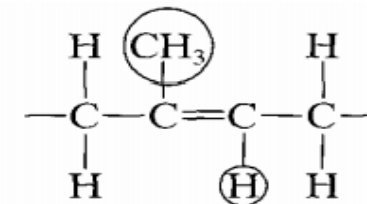


Geometrical isomerism

Geometrical isomerism: consider two carbon atoms bonded by a double bond in a chain. H atom or radical R bonded to these two atoms can be on the same side of the chain (cis structure) or on opposite sides of the chain (trans structure).

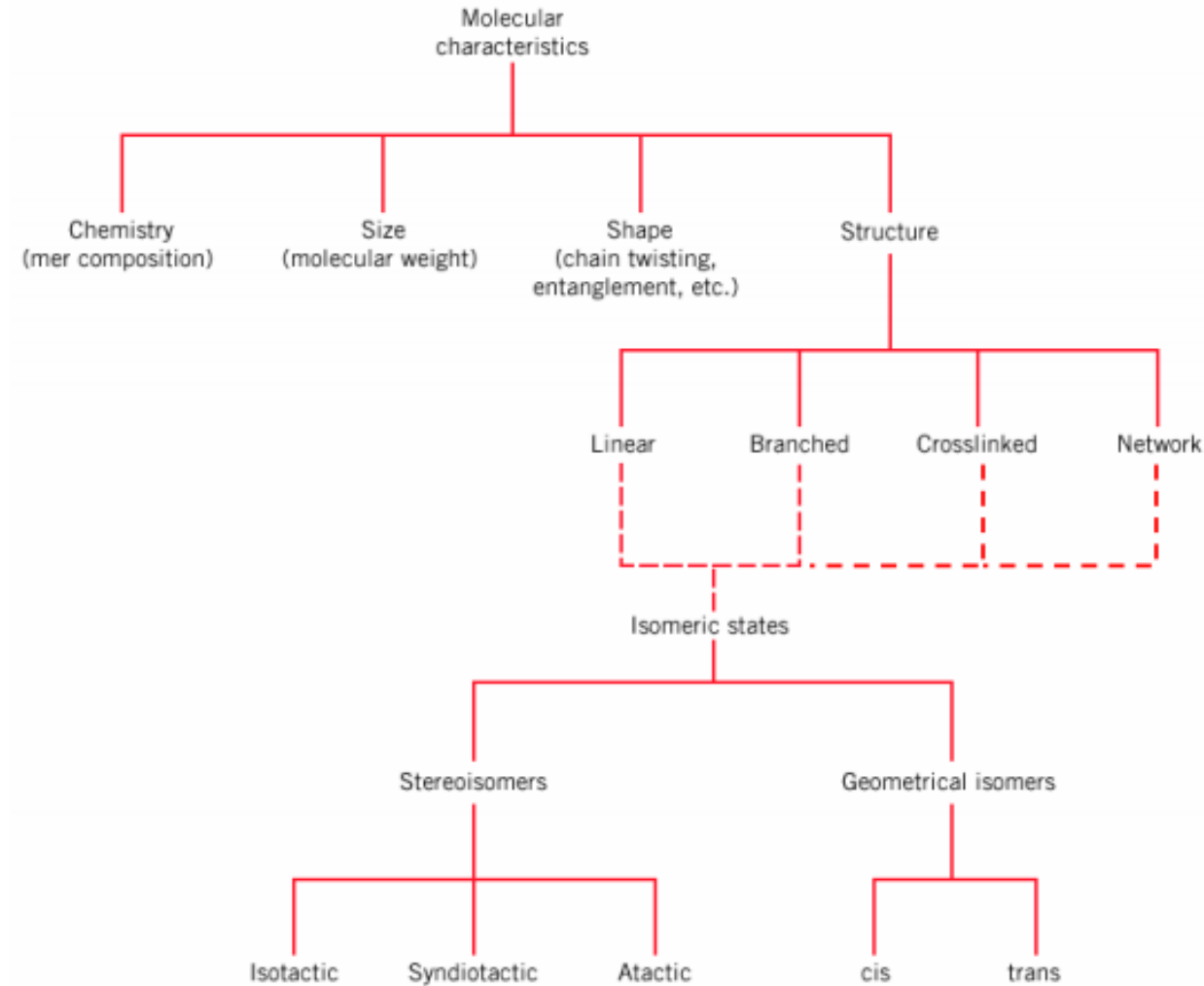


Cis-polyisoprene



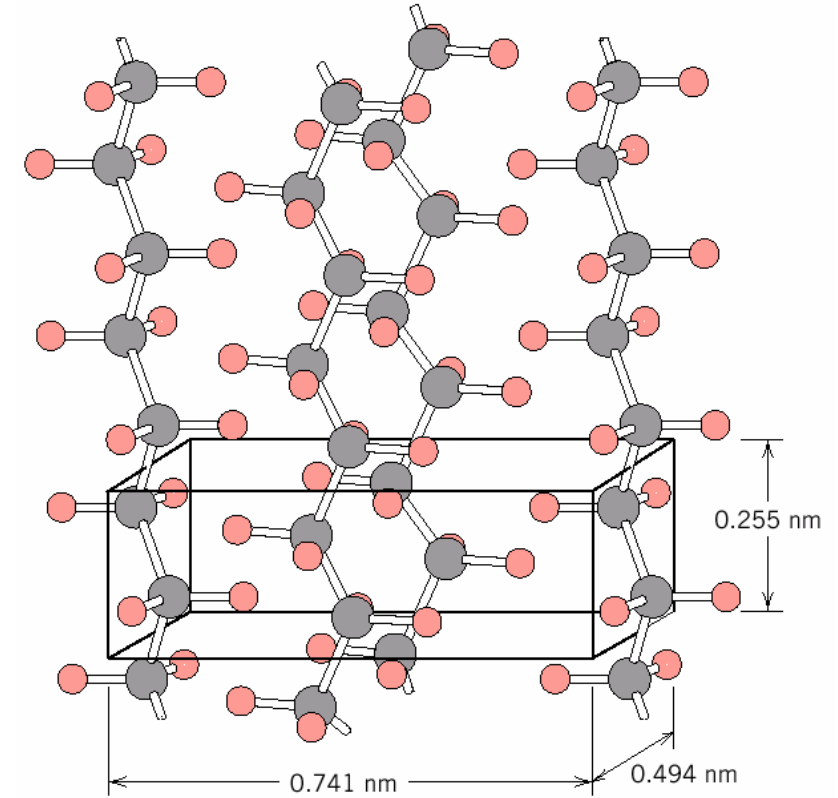
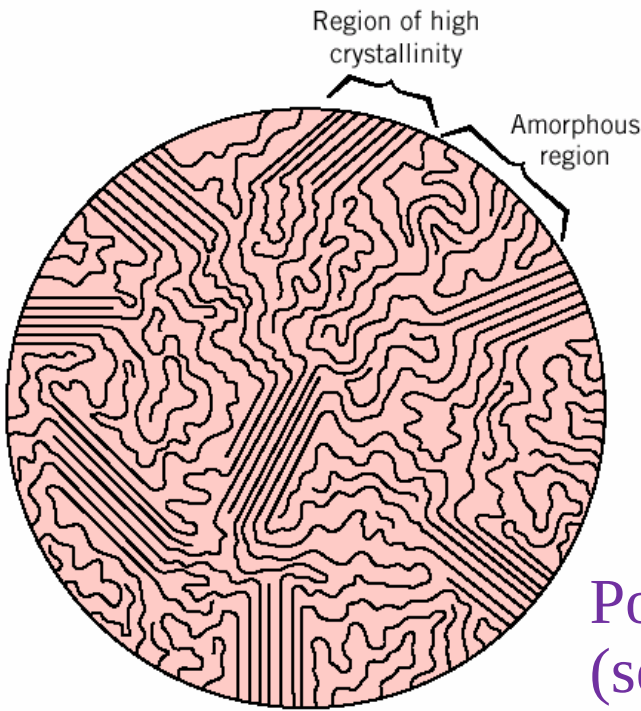
Trans-polyisoprene

Size – Shape – Structure classification



Polymer crystallinity

Atomic arrangement in polymer crystals is more complex than in metals or ceramics (unit cells are typically large and complex).



Polyethylene

Polymer molecules are often partially crystalline (semicrystalline), with crystalline regions dispersed within amorphous material

Polymer Crystallinity (II)

Degree of crystallinity is determined by

- **Rate of cooling during solidification:** time is necessary for chains to move and align into a crystal structure
- **Mer complexity:** crystallization less likely in complex structures, simple polymers, such as polyethylene, crystallize relatively easily
- **Chain configuration:** linear polymers crystallize relatively easily, branches inhibit crystallization, network polymers almost completely amorphous, crosslinked polymers can be both crystalline and amorphous
- **Isomerism:** isotactic, syndiotactic polymers crystallize relatively easily - geometrical regularity allows chains to fit together, atactic polymers is difficult to crystallize
- **Co-polymerism:** easier to crystallize if mer arrangements are more regular - alternating, block can crystallize more easily as compared to random and graft

More crystallinity: higher density, more strength, higher resistance to dissolution and softening by heating

Polymer Crystallinity (III)

Crystalline polymers are denser than amorphous polymers, so the degree of crystallinity can be obtained from the measurement of density:

$$\% \text{ crystallinity} = \frac{\rho_c(\rho_s - \rho_a)}{\rho_s(\rho_c - \rho_a)} \times 100$$

ρ_c : Density of perfect crystalline polymer

ρ_a : Density of completely amorphous polymer

ρ_s : Density of partially crystalline polymer that we are analyzing

